

HPC Allocation Request for Quantum-Active Learning DFT

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Program: Generative Quantum Machine Learning for Cost-Efficient, High-Strength Materials
Discovery

Executive Summary

The quantum-active learning (QAL) workflow couples quantum support vector/gaussian process regressors with density-functional theory (DFT) feedback to screen and validate high-entropy alloys. Closed-loop iterations require three production Quantum ESPRESSO (QE) job classes—`scf_screen`, `vc_relax`, and `elastic_eval`—that were benchmarked previously. The measured footprints (Tables 1 and 2) motivate a request for:

- **Compute:** 1,200 CPU core-hours on CPU-only nodes (rounded up from 1,022 core-hours to cover scheduling overhead and reruns), provisioned on 32-core nodes with ≥ 120 GiB memory.
- **Storage:** 60 GiB of project space plus 250 GiB of scratch to host QE temporary files, elastic-strain wavefunctions, and MLflow artifacts (benchmark workloads require 42.3 GiB including safety buffers).
- **Queue policy:** Job arrays of width 16–24 to keep the active-learning loop saturated while respecting per-partition limits.

I have all artifacts required for immediate deployment: benchmark summaries, storage forecasts, SLURM templates, and environment manifests.

Workload Overview

- `scf_screen`: Fast single-point SCF solves that supply acquisition functions with updated energetics during each QAL iteration.
- `vc_relax`: Variable-cell relaxations for shortlisted alloys prior to downstream mechanical evaluation; captures lattice and ionic rearrangements required for manuscript-quality claims.
- `elastic_eval`: Finite-strain elasticity batches that deliver elastic tensors, yield proxies, and uncertainty annotations for the PDA milestone gate.

Each profile was executed with QE 7.4.1, SSPP PBE 1.3 pseudopotentials, and the orchestration scripts `scripts/dft/qe_benchmark.py` and `scripts/dft/storage_planner.py`.

Benchmark Evidence

Per-run measurements (Table 1) are sourced directly from the benchmark summaries under `data/dft_workflow/<run>`. The aggregated storage and runtime projections (Table 2) follow `docs/hpc/storage_forecast_plan.json`, which incorporates the safety factors indicated.

Table 1: Benchmark-verified per-run resource footprint (Task T5R.1).

Job class	Runs $\times$ safety	Wall (s)	CPU (s)	Peak RSS (GiB)	Disk/run (GiB)
<code>scf.screen</code>	200×1.5	13.34	13.09	0.19	0.016 6
<code>vc.relax</code>	60×1.3	641.88	628.04	0.26	0.016 8
<code>elastic.eval</code>	24×1.2	2 183.62	2 170.84	0.31	1.249 4

Table 2: Scaled allocation request derived from benchmarking and storage forecasts. Core-hours assume the indicated MPI task counts.

Job class	Total wall (hr)	Planned MPI tasks	Core-hours	Total disk (GiB)
<code>scf.screen</code>	1.11	16	17.79	4.98
<code>vc.relax</code>	13.91	32	445.04	1.31
<code>elastic.eval</code>	17.47	32	559.01	35.98
Total	32.49	—	1 021.84	42.27

Resource Request Rationale

- **Compute scale:** The elastic evaluation workload dominates at 559 core-hours when executed with 32 MPI ranks; the relaxation stage contributes another 445 core-hours. Rounding the total (1,021.8 core-hours) to 1,200 core-hours provides a 15% contingency for reruns, parameter sweeps, and prospective feature-map ablations required by the PDA milestone process.
- **Concurrency:** Active learning requires multiple DFT callbacks per iteration. Queueing `vc.relax` jobs in arrays of 8 and `elastic.eval` batches in arrays of 4 keeps total wall-clock under two days for each discovery cycle while remaining under typical 32-job array limits.
- **Memory headroom:** Although peak RSS remained below 0.31 GiB per MPI rank, reserving 120 GiB per node allows for higher plane-wave cutoffs (observed up to 650 Ry) and larger supercells anticipated for HEA mixing studies without resubmitting allocation amendments.
- **Storage and scratch:** Elastic evaluations emit ≈ 1.25 GiB per run. The aggregated projection of 42.3 GiB (with safety factors) motivates a 60 GiB persistent allocation. Requesting 250 GiB of scratch accommodates simultaneous MPI jobs and transient `qe.tmp` directories emphasized in `docs/hpc/qe_slurm_template.sh`.
- **Operational readiness:** The SLURM template, environment manifest, and MLflow tracking already align with the target HPC stack, minimizing bring-up risk. Storage forecasts and benchmark artifacts deliver the traceability required by RKMA.

Execution Plan and Risk Mitigation

- **Job staging:** Use SLURM job arrays driven by `scripts/dft/storage_planner.py` to batch candidates per acquisition iteration. The planner guarantees scratch usage stays within allocation limits.
- **Monitoring:** MLflow logs (metrics and artifacts) and the QE output archives will be mirrored nightly to object storage, satisfying RKMA reproducibility checks and enabling quick rollback

on failure.

- **Fallback:** Should queue wait times exceed PDA-defined SLAs, short `scf_screen` jobs can be diverted to on-prem nodes; the HPC allocation is prioritized for `vc_relax` and `elastic_eval`, which cannot be executed economically on workstations.
- **Contingency:** The 15% compute margin covers reruns triggered by pseudopotential updates, DFT convergence restarts, or acquisition function perturbations introduced during robustness studies led by BRA.

Summary

Benchmark data confirms that the QAL pipeline will consume approximately 1,022 core-hours and 42 GiB of output storage across the three production QE workloads. A request for 1,200 CPU core-hours, 60 GiB project storage, and 250 GiB scratch—all on CPU-only nodes compatible with QE 7.4.1—provides the capacity needed to execute upcoming active-learning campaigns, satisfy PDA milestone gates, and maintain reproducible artefact management. Approval will unblock the transition from local benchmarking to full-scale HPC-backed discovery.